Generative-AI Prompt-Engineered Cyberbullying Detection

Reasoning and Classification of Cyberbullying Tweets Using Prompt Engineering and a Rule-Based Hybrid Scoring Layer

A Technical Project Report

Detailed Experimental Design, System Implementation, and Results

Generative Model: **Groq llama-3.1-8b-instant** | Approach: **training-free, prompt-only**

Dataset: 47,692 labelled tweets → 46,016 unique | 6 balanced classes

Best system: **Exp 7 – Hybrid (Reasoning+Rules)** — Accuracy 0.443, Macro-F1 0.404, Binary Recall 0.878

Abstract

Cyberbullying is a pervasive harm on social media, and automated moderation systems frequently miss the context, intent, and tone that distinguish abusive speech from ordinary negativity or banter. This project asks whether a Generative-AI model, guided *only* through prompt engineering and supported by a lightweight rule-based layer, can both classify and *explain* cyberbullying without any model training. A publicly available Twitter corpus of 47,692 labelled tweets (46,016 after de-duplication, six balanced classes) is used not as training data but as a test bench against which the model is evaluated, in the same way one might give a human reviewer many case examples and observe their judgements.

Five prompting strategies — zero-shot, few-shot, chain-of-thought, persona, and a few-shot+chain-of-thought combination — are benchmarked on a stratified 600-tweet evaluation sample. A transparent toxicity scorer built from category-grouped lexicons and sentiment analysis serves as a non-AI baseline and as a safety layer. The two paradigms are then fused into a hybrid 'Reasoning + Rules' system. The hybrid achieves the best results (accuracy 0.443, macro-F1 0.404) while retaining the language model's high binary recall (0.878). The central methodological finding is that the productive way to combine the two paradigms is *category arbitration* — using the lexicon to correct the model's category when an explicit marker fires — rather than a recall safety-net, because the model's residual errors are implicit cases that the lexicon also cannot see. The report documents the full experimental design, a reproducible nine-stage implementation, and per-experiment results.

Table of Contents

Chapter 1. Introduction

1.1 Background and Motivation

Cyberbullying today is not confined to a single platform or moment; it follows people across short posts, memes, and casual comments, and a few words written in anger or sarcasm can leave a lasting emotional scar. Platforms attempt to detect harmful content with machine-learned systems, but these often fail to grasp context, intent, or tone. The very same phrase can be harmless banter among friends yet deeply offensive when aimed at a person because of their age, religion, gender, or ethnicity. This gap between human understanding and automated moderation is the motivation for the present work.

1.2 Problem Statement

Conventional approaches train supervised machine-learning or deep-learning classifiers, which demand labelled data, tuning, and significant compute, and which typically emit only a class label with no human-readable justification. The problem addressed here is twofold: (i) can a Generative-AI model classify cyberbullying accurately *without training*, purely through structured prompting; and (ii) can it simultaneously produce a transparent explanation that a teacher, counsellor, or moderator can trust and act upon?

1.3 Research Gaps

- Most cyberbullying studies rely on supervised models that require training and tuning; little work explores purely prompt-based generative classifiers that avoid training entirely.
- Existing systems usually output only a label, with no meaningful explanation; there is a clear gap in transparent, reasoning-focused detectors.
- Hybrid decision mechanisms, in which LLM reasoning is cross-checked by simple lexical and sentiment scores, are rarely studied as a way to reduce error and hallucination.
- Few works systematically compare prompting strategies (zero-shot, few-shot, chain-of-thought, persona) for nuanced categories such as age, ethnicity, gender, and religion.

1.4 Aim and Objectives

The aim is to build and evaluate a lightweight, explainable, training-free cyberbullying detection pipeline. The specific objectives are:

- To design and benchmark multiple prompting strategies for six-way cyberbullying classification.
- To construct a transparent rule-based toxicity and sentiment scorer as a baseline and safety layer.
- To fuse generative reasoning with the rule layer into a hybrid system and measure its benefit.
- To evaluate explanation quality, not only label accuracy, as a first-class objective.
- To document where the system struggles (sarcasm, coded language, cultural phrasing).

1.5 Novelty and Contributions

- A **training-free** cyberbullying detection pipeline using only LLM prompting plus rule-based scoring, suited to low-compute settings.
- A hybrid **‘Reasoning + Rules’** framework whose fusion mechanism — category arbitration — is derived empirically from where each paradigm is strong.

- **Explainability as a first-class objective**, with structured justifications and an LLM-as-judge evaluation of explanation quality.
- A **prompt-level benchmark** quantifying how zero-shot, few-shot, chain-of-thought, and persona prompting compare on fine-grained cyberbullying categories.

1.6 Scope

The study uses a single open Twitter corpus and a single small, fast, openly-served model (Groq llama-3.1-8b-instant). It evaluates classification and explanation on a balanced 600-tweet sample, with the free rule layer additionally applied to all 46,016 unique tweets. Real-time deployment, multilingual handling, and human-subject studies are out of scope.

Chapter 2. Dataset and Experimental Design

2.1 Dataset

The corpus contains 47,692 tweets, each labelled with one of six cyberbullying types. After removing 1,676 exact-duplicate tweets, 46,016 unique tweets remain. The classes are well balanced (Table 1), which makes accuracy and macro-averaged F1 directly meaningful and avoids the majority-class bias that distorts imbalanced corpora.

Class	Tweets (after de-duplication)
religion	7,995
age	7,991
ethnicity	7,952
not_cyberbullying	7,937
gender	7,898
other_cyberbullying	6,243
Total (unique)	46,016

Table 1. Class distribution of the cleaned dataset.

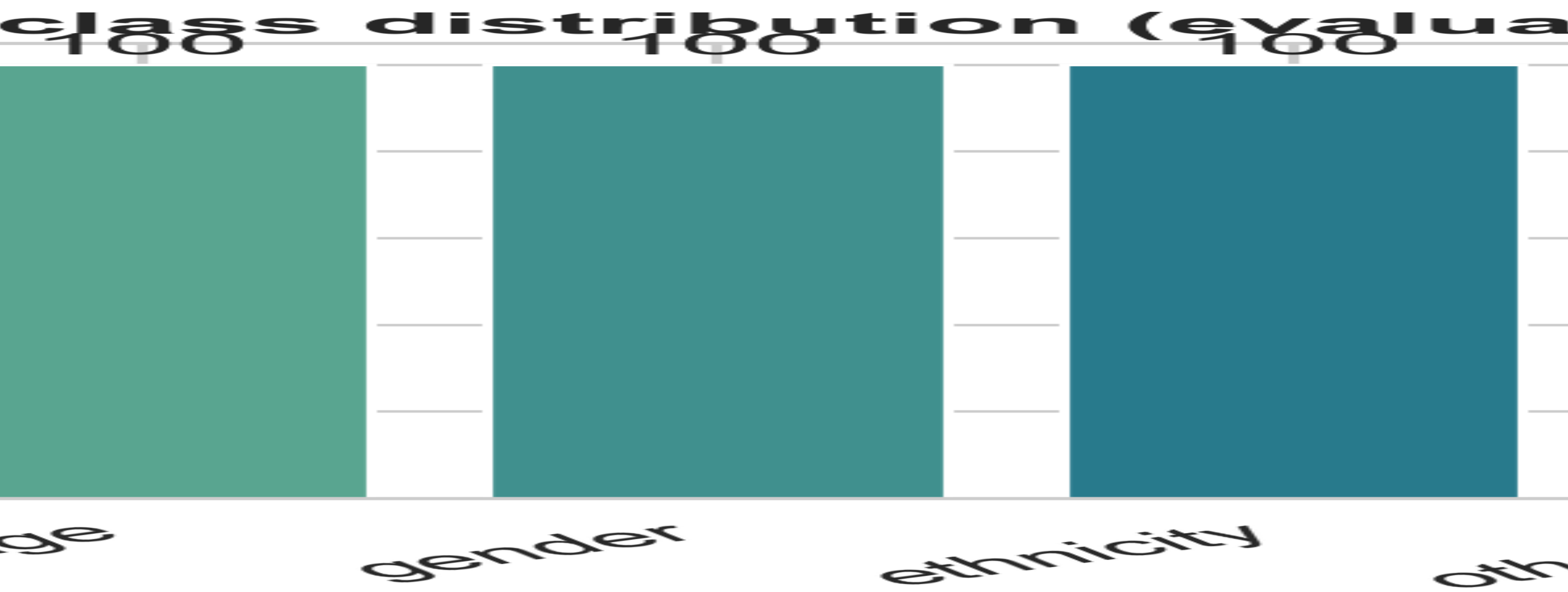

Figure 1. Class distribution of the cleaned dataset.

2.2 Preprocessing

Cleaning is deliberately light, because a language model benefits from seeing a tweet much as a human would. Two text columns are retained: the original tweet (fed to the model) and a normalised version (used by the rule layer) in which HTML entities are unescaped, URLs and @-mentions are removed, and the hash symbol is stripped while the hashtag word is kept. Exact duplicates are removed, and a fixed random seed governs all sampling so the experiments are reproducible.

2.3 Evaluation Protocol

The design is training-free: the labelled data is a test bench, never a training set. Each experiment follows one loop — read a tweet, build a prompt, call the model, parse a strict JSON answer, and compare the predicted label with the gold label. Because sending 46,016 tweets to a model is costly and slow, the model experiments use a **stratified sample of 600 tweets (100 per class)**; the free rule layer additionally runs on the full dataset. The decoding temperature is fixed at zero, and every prompt–response pair is cached so re-runs are instant and identical.

2.4 Structured Output Schema

To make every strategy directly comparable and every decision explainable, all prompts force the same JSON object:

```
{ "label", "target_group", "intent", "intensity", "explanation" }
```

The *label* is one of the six classes; *target_group* names who is attacked; *intent* and *intensity* capture severity; and *explanation* is a one-sentence justification. This schema operationalises explainability and yields machine-readable output for automatic scoring.

2.5 Prompting Strategies

Five prompting styles are studied. They share the identical task description and output schema; only the way the model is asked to think changes, which isolates the effect of prompt design.

- **Zero-shot** — a plain instruction with no examples or reasoning.
- **Few-shot** — two to three hand-written, synthetic labelled examples per category (no dataset tweets, hence no leakage).
- **Chain-of-Thought** — an explicit four-step reasoning instruction before the answer.
- **Persona** — the model is cast as an experienced content-moderation officer who also advises school counsellors.
- **Few-shot + CoT** — examples and reasoning combined.

2.6 Experiment Catalogue

#	Experiment	Purpose
1	Rule-based baseline	Lexicon + sentiment toxicity scorer; the non-AI floor and the safety layer.
2	Zero-shot	Pure generative ability with no scaffolding.
3	Few-shot	Effect of in-context examples.
4	Chain-of-Thought	Effect of explicit step-by-step reasoning.
5	Persona	Effect of role-conditioning.
6	Few-shot + CoT	Whether examples and reasoning stack.
7	Hybrid (Reasoning+Rules)	Fusion of the best model with the lexicon.
8	Robustness / hard cases	Behaviour on sarcasm, coded, and quoted tweets.
9	Explanation quality	LLM-as-judge rubric scoring of justifications.
10	Benchmark synthesis	Roll-up tables, ablation, and reporting.

Table 2. The ten experiments and their purpose.

2.7 Evaluation Metrics

- **Accuracy** and **macro-F1** on the six-class task (macro-F1 is the primary metric, since classes are balanced).
- **Per-class F1**, to reveal which categories are confused.
- **Binary precision and recall** (bully vs not), the metric most relevant to moderation safety.
- **Parse-OK rate**, the fraction of replies returning valid JSON.
- **Explanation quality**, a composite of four rubric checks (Experiment 9).

Chapter 3. System Implementation

3.1 Architecture and Pipeline

The system is implemented as a reproducible nine-stage pipeline. Each stage is an independent script that writes its outputs to disk, so stages can be executed and inspected one at a time and the whole study can be replayed from any point.

Stage	Module	Responsibility
00	check_api_keys	Health-checks every API key and records the working subset, so restricted keys are skipped.
01	prepare_data	Cleans, de-duplicates, and builds the stratified 600-tweet evaluation sample.
02	score_lexicon	Rule-based toxicity and sentiment scorer (Experiment 1) on sample and full data.
03	run_prompts	Runs the five prompt styles (Experiments 2–6) over the sample.
04	fuse_hybrid	Builds the hybrid (Experiment 7) by category arbitration.
05	evaluate_metrics	Computes metrics, confusion matrices, and figures.
06	analyze_hard_cases	Robustness slice (Experiment 8) and failure gallery.
07	judge_explanations	LLM-as-judge explanation scoring (Experiment 9).
08 / 09	build_report / build_pdf	Synthesis (Experiment 10) and this document.

Table 3. The nine-stage implementation pipeline.

3.2 Engineering Details

3.2.1 API-key rotation and resilience

Thirteen API keys were supplied; a preflight health-check found nine live and four belonging to restricted organisations. The client round-robins across the live keys on every call so that no single key exceeds the free-tier rate limit, and it permanently drops any key that returns a restricted or invalid error at runtime. This allowed a batch of roughly three thousand calls to complete without manual intervention.

3.2.2 Caching and reproducibility

Every (model, prompt) pair is hashed and its response cached as JSON. Re-running an experiment therefore costs nothing and returns identical results, which is essential for a study that compares prompt variants.

3.2.3 Robust JSON parsing

The first JSON object in each reply is extracted and, if necessary, repaired (for example, trailing commas are removed). Predicted labels are then normalised onto the six canonical classes, mapping common variants such as ‘racism’ to ethnicity. Across all three thousand model calls the parse-OK rate was 100%.

3.2.4 Concurrency

Eight worker threads (kept below the nine-key count) issue calls concurrently, reducing the full five-style run to roughly twenty minutes.

3.3 Implementation of Each Experiment

Each experiment is described below by its objective, its design rationale, the concrete implementation steps, and a short note on outputs.

3.3.1 Experiment 1 — Rule-based Baseline (Safety Layer)

Objective. To establish a transparent, training-free, non-AI floor and to provide a lexical safety signal that the hybrid can later use.

Design rationale. A compact, fully auditable lexicon is preferable to an opaque toxicity model: every decision can be traced to a specific term, which serves the explainability aim and keeps compute negligible.

Implementation steps.

Step 1. Load the cleaned tweets together with the normalised text column.

Step 2. Define compact, human-readable lexicons grouped by target category (ethnicity, religion, gender, age, other).

Step 3. Define a small hard-flag set (explicit slurs and phrases such as 'kill yourself') that almost always indicate bullying.

Step 4. For each tweet, count lexicon hits per category and compute a VADER sentiment compound score.

Step 5. Combine into a toxicity score (0.7 × lexicon density + 0.3 × negativity), forced to at least 0.9 if a hard flag fires.

Step 6. Predict the category as the lexicon group with the most hits, and mark the tweet as bullying when toxicity reaches the 0.35 threshold.

Step 7. Map the decision into the six-class space (the category guess if bullying, otherwise not_cyberbullying).

Step 8. Write per-tweet scores for both the 600-tweet sample and the full dataset, and report binary accuracy.

Outputs. Outputs the scored sample (consumed by the hybrid) and the full-data baseline. Binary accuracy on the full corpus is 0.511, confirming that a small lexicon catches explicit abuse but misses much implicit bullying.

3.3.2 Experiment 2 — Zero-shot

Objective. To measure the model's raw classification ability with no scaffolding.

Design rationale. Zero-shot is the natural baseline against which every richer prompt is compared; any gain from examples or reasoning must be measured relative to it.

Implementation steps.

Step 1. Build a base task string listing the six categories with short descriptions.

Step 2. Append the strict JSON output contract so the reply is machine-parseable.

Step 3. Set a minimal system prompt ('You are a precise text classifier').

Step 4. Insert only the target tweet, with no examples and no reasoning instruction.

Step 5. Send the prompt through the cached, key-rotated client at temperature zero.

Step 6. Extract and parse the first JSON object from the reply.

Step 7. Normalise the predicted label onto one of the six canonical classes.

Step 8. Persist identifier, gold label, prediction, target group, intent, intensity, explanation, and parse status.

Step 9. Score accuracy and macro-F1 against the gold labels.

Outputs. Achieves accuracy 0.383 and macro-F1 0.363 — a surprisingly strong floor for a small model on a six-way task.

3.3.3 Experiment 3 — Few-shot

Objective. To test whether a few in-context examples improve fine-grained classification.

Design rationale. Examples can teach the label boundaries (for instance, religion versus ethnicity) without any training; the examples are synthetic to eliminate any risk of leaking evaluation tweets.

Implementation steps.

Step 1. Reuse the base task and JSON contract from Experiment 2.

Step 2. Hand-write one clear, synthetic example per category.

Step 3. Format the examples as Tweet/Answer pairs and prepend them to the prompt.

Step 4. Append the target tweet after the examples.

Step 5. Call the model with identical client settings for comparability.

Step 6. Parse and normalise the predicted label.

Step 7. Persist predictions and explanations.

Step 8. Compare macro-F1 against zero-shot to isolate the effect of examples.

Outputs. Macro-F1 rises slightly to 0.378 (the best macro-F1 among the pure prompts), though six-class accuracy dips to 0.365, indicating examples sharpen some categories while biasing others.

3.3.4 Experiment 4 — Chain-of-Thought

Objective. To test whether forcing explicit reasoning improves judgement.

Design rationale. Step-by-step reasoning is widely reported to help larger models on complex tasks; this experiment asks whether the benefit holds for a small model on nuanced categories.

Implementation steps.

Step 1. Reuse the base task and JSON contract.

Step 2. Add a four-step reasoning instruction: identify the target, analyse tone, judge intent, then choose the trait.

Step 3. Instruct the model to reason internally and output only the final JSON.

Step 4. Use a reasoning-oriented system prompt.

Step 5. Send the target tweet and collect the reply.

Step 6. Parse the JSON label and the structured fields.

Step 7. Normalise and store the predictions.

Step 8. Evaluate and compare against zero-shot.

Outputs. Accuracy falls to 0.353 and macro-F1 to 0.333; on this small model the reasoning scaffold does not help and slightly hurts.

3.3.5 Experiment 5 — Persona

Objective. To test whether role-conditioning changes strictness and judgement.

Design rationale. Casting the model as a fair, context-aware moderator may align its behaviour with the task without any examples or explicit reasoning steps.

Implementation steps.

Step 1. Reuse the base task and JSON contract.

Step 2. Write a rich persona system prompt: an experienced moderation officer who advises school counsellors, fair and protective of vulnerable groups.

Step 3. Keep the user prompt simple (‘As the moderator, classify this tweet’).

Step 4. Call the model with the persona system message.

Step 5. Parse and normalise the predicted label.

Step 6. Persist predictions and the (often more careful) explanations.

Step 7. Score accuracy and macro-F1.

Step 8. Compare against the other styles.

Outputs. The strongest pure prompt by accuracy (0.400), with the highest binary recall (0.878); it becomes the base for the hybrid.

3.3.6 Experiment 6 — Few-shot + CoT

Objective. To test whether examples and reasoning combine additively.

Design rationale. If both techniques help independently, combining them might compound the gains; this experiment checks that assumption.

Implementation steps.

Step 1. Reuse the base task and JSON contract.

Step 2. Prepend the hand-written few-shot examples.

Step 3. Add the four-step reasoning instruction.

Step 4. Use the reasoning-oriented system prompt.

Step 5. Send the target tweet and collect the reply.

Step 6. Parse, normalise, and store the predictions.

Step 7. Evaluate and compare against the single-technique prompts.

Outputs. The weakest configuration (accuracy 0.322, macro-F1 0.327): the two techniques interfere rather than stack on this model.

3.3.7 Experiment 7 — Hybrid (Reasoning + Rules)

Objective. To combine the model's reasoning with the lexicon so that the whole exceeds either part.

Design rationale. Diagnostic analysis showed the model has high recall but confuses categories, whereas the lexicon has low recall but names the category precisely when an explicit marker fires. The productive fusion is therefore category arbitration, not a recall safety-net, because the model's misses are implicit cases (mean toxicity 0.04) the lexicon also cannot see.

Implementation steps.

Step 1. Choose the best pure model run (persona) as the base prediction.

Step 2. Join the model predictions with the Experiment 1 rule scores on tweet identifier.

Step 3. Diagnose where each method is strong (model recall vs lexical category precision).

Step 4. Verify on disagreement cases that the fired rule category is correct far more often than the model (45 versus 8).

Step 5. Define category arbitration: when the lexicon fires a confident category and toxicity reaches 0.5, override the model's category with the lexicon's.

Step 6. Also record a recall safety-net and a binary-union strategy for comparison.

Step 7. Apply arbitration to produce the hybrid prediction.

Step 8. Report rescued versus broken cases and the accuracy gain.

Step 9. Evaluate against all other systems.

Outputs. Arbitration applied 48 overrides — 33 correct rescues against 7 breaks — lifting accuracy from 0.400 to 0.443 and macro-F1 to 0.404, the best of any system, while preserving recall at 0.878.

3.3.8 Experiment 8 — Robustness on Hard Cases

Objective. To document where the system struggles: sarcasm, coded language, quoted irony, and reclaimed terms.

Design rationale. Aggregate accuracy can hide systematic blind spots; isolating hard tweets exposes them and produces concrete error examples for qualitative discussion.

Implementation steps.

Step 1. Reuse the predictions already computed for Experiments 2–6 (no new calls).

Step 2. Define a 'hard' heuristic from sarcasm markers, emojis, quoted irony, and reclaimed in-group terms.

Step 3. Flag each evaluation tweet as hard or easy.

Step 4. Compute per-style accuracy on the hard, easy, and overall subsets.

Step 5. Collect the misclassified hard cases into a failure gallery.

Step 6. Save the summary and gallery for analysis.

Outputs. Persona is the most robust on hard tweets; few-shot+CoT degrades most. Seventy-nine hard errors are catalogued for inspection.

3.3.9 Experiment 9 — Explanation Quality

Objective. To evaluate the *quality* of justifications, not just label correctness.

Design rationale. Because explainability is a primary objective, the explanations must be assessed directly; an LLM-as-judge applies a consistent rubric at scale.

Implementation steps.

Step 1. Take the best style's predictions and explanations (persona).

Step 2. Draw a stratified subset of 20 tweets per class (120 explanations).

Step 3. Define a four-point rubric: names the target, cites the reason, intent coherent, consistent with the gold label.

Step 4. Prompt a judge model to score each explanation as 0/1 flags in JSON.

Step 5. Run the judge concurrently through the cached client.

Step 6. Average the flags into a composite quality score.

Step 7. Aggregate overall and per-class explanation quality.

Outputs. Composite quality is 0.569: intent is described very coherently (0.95) but the precise target group is often not named (0.31).

3.3.10 Experiment 10 — Benchmark Synthesis

Objective. To consolidate all results into comparison tables, an ablation, and reports.

Design rationale. A single synthesis stage guarantees the headline numbers are computed once from the same source and reused consistently across the Markdown report and this document.

Implementation steps.

Step 1. Load the master metrics table from the evaluation stage.

Step 2. Build the six-class comparison and the prompt-style ablation (delta versus zero-shot).

Step 3. Quantify the hybrid's improvement over the best pure model.

Step 4. Fold in the hard-case and explanation-quality results.

Step 5. Emit the Markdown report and this PDF.

Outputs. Produces the artifacts presented in Chapter 4.

Chapter 4. Results and Discussion

4.1 Master Comparison

System	Acc.	Macro-F1	Wt-F1	Bin. Recall	Parse-OK
Exp 7 – Hybrid (Reasoning+Rules)	**0.443**	**0.404**	**0.404**	**0.878**	**1.00**
Exp 3 – Few-shot	0.365	0.378	0.378	0.838	1.00
Exp 1 – Rule-based (no LLM)	0.417	0.371	0.371	0.396	n/a
Exp 5 – Persona	0.400	0.364	0.364	0.878	1.00
Exp 2 – Zero-shot	0.383	0.363	0.363	0.838	1.00
Exp 4 – Chain-of-Thought	0.353	0.333	0.333	0.868	1.00
Exp 6 – Few-shot + CoT	0.322	0.327	0.327	0.872	1.00

Table 4. Six-class comparison of all systems, sorted by macro-F1.

The hybrid leads on both accuracy and macro-F1 while retaining the model's high binary recall. The rule baseline reaches competitive accuracy with extremely high precision (0.99) but very low recall (0.40): it is correct when it fires, but it fires only on explicit slurs. The language models, in contrast, recover most bullying (recall 0.84–0.88) but confuse the fine-grained categories.

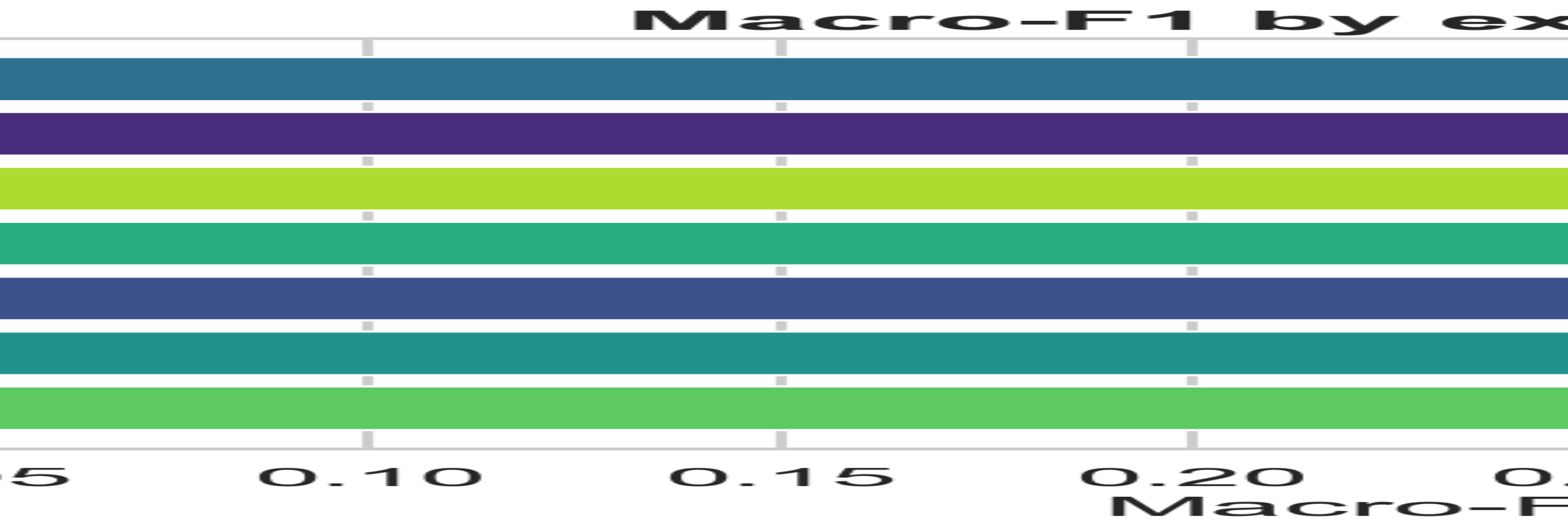

Figure 2. Macro-F1 by experiment.

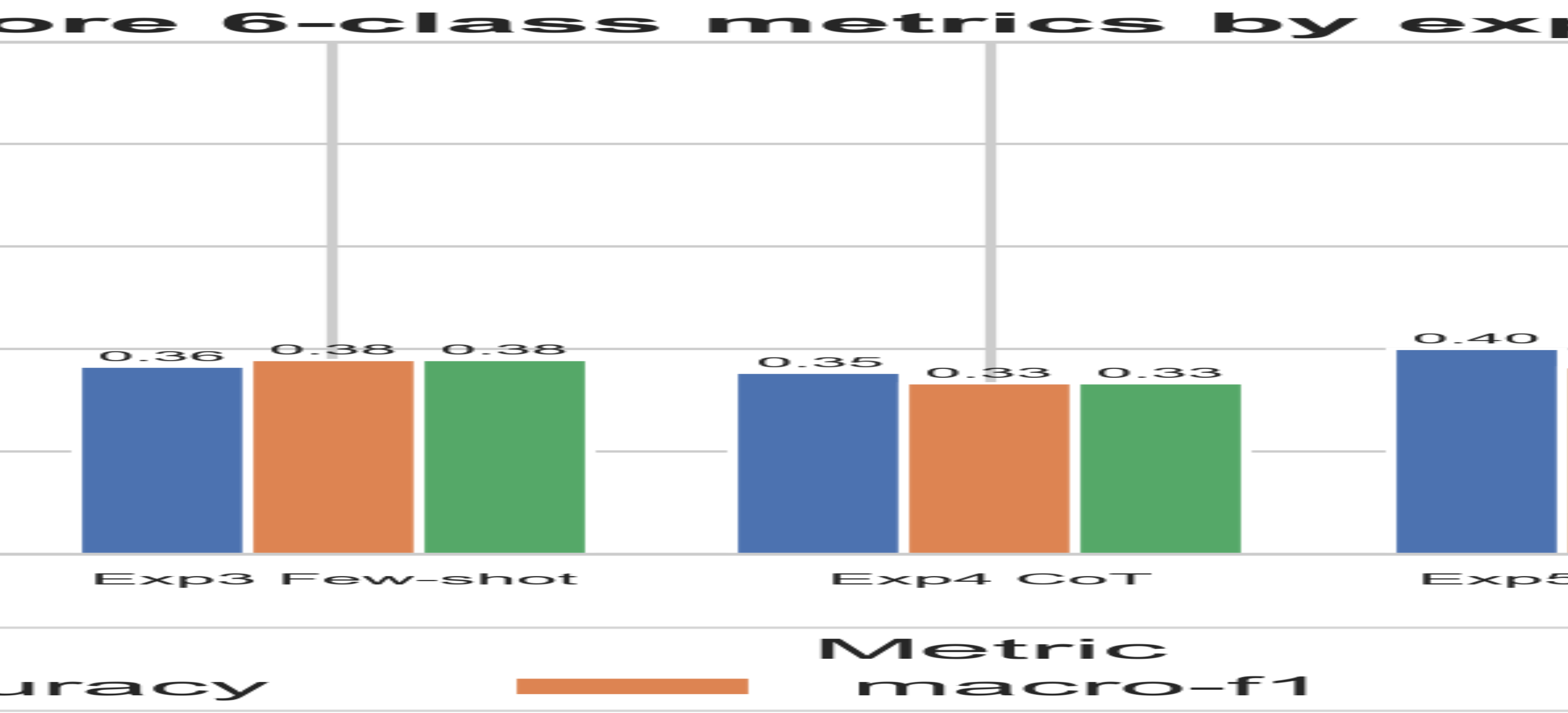

Figure 3. Accuracy, macro-F1, and weighted-F1 across systems.

4.2 Prompt-Style Ablation

Prompt style	Accuracy	Macro-F1	∆ Macro-F1 vs zero-shot
Exp 2 – Zero-shot	0.3833	0.3626	+0.0000
Exp 3 – Few-shot	0.3650	0.3782	+0.0156
Exp 4 – Chain-of-Thought	0.3533	0.3334	-0.0292
Exp 5 – Persona	0.4000	0.3643	+0.0017
Exp 6 – Few-shot + CoT	0.3217	0.3265	-0.0361

Table 5. Prompt-style ablation relative to the zero-shot baseline.

Few-shot gives a small positive change in macro-F1, persona is essentially neutral on macro-F1 but best on accuracy, and both reasoning configurations reduce performance. The clear lesson is that, for a small model on this fine-grained task, **role-conditioning is more effective than reasoning scaffolds**, and stacking techniques can be counter-productive.

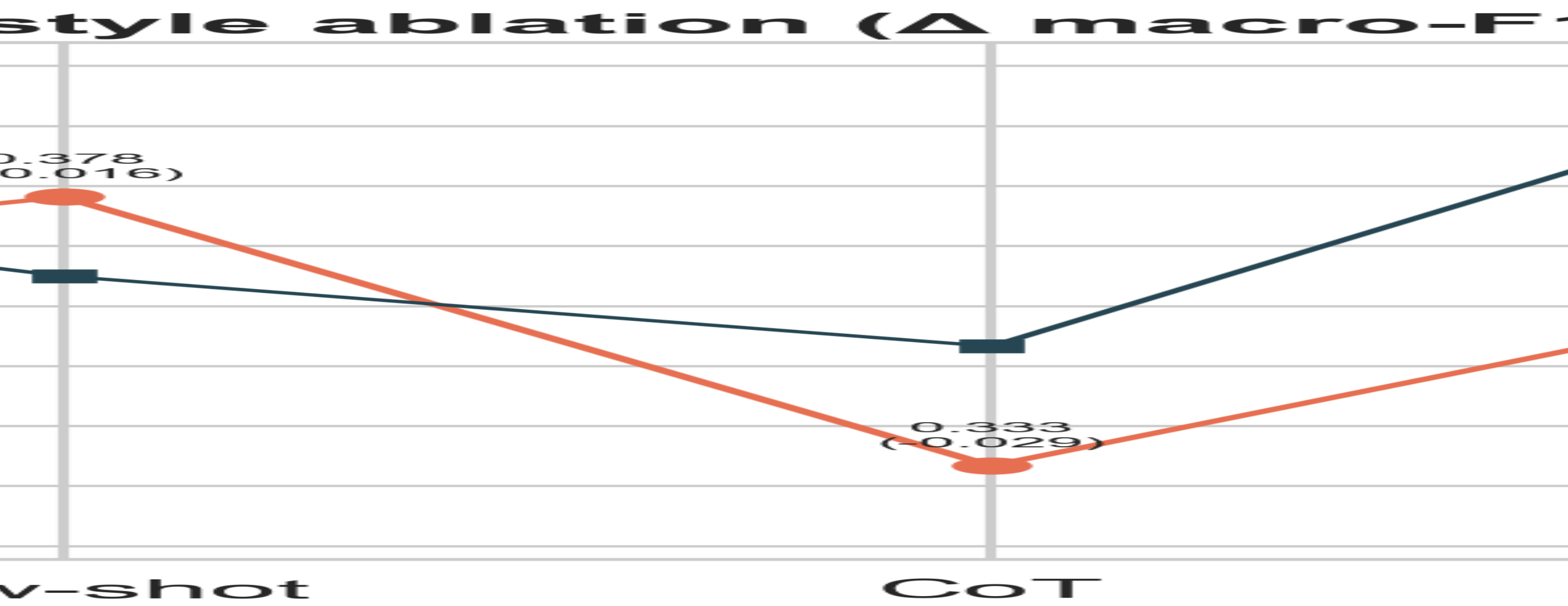

Figure 4. Prompt-style ablation.

4.3 Per-Class Analysis

System	relig.	age	gender	ethnic.	other	not_cb
Exp 7 – Hybrid (Reasoning+Rules)	0.28	0.00	0.72	0.69	0.27	0.46
Exp 3 – Few-shot	0.37	0.08	0.50	0.63	0.21	0.49
Exp 1 – Rule-based (no LLM)	0.26	0.00	0.75	0.78	0.04	0.39
Exp 5 – Persona	0.18	0.00	0.66	0.63	0.25	0.46
Exp 2 – Zero-shot	0.21	0.00	0.59	0.66	0.24	0.47
Exp 4 – Chain-of-Thought	0.10	0.00	0.56	0.66	0.22	0.46
Exp 6 – Few-shot + CoT	0.30	0.00	0.38	0.58	0.21	0.48

Table 6. Per-class F1 for every system.

The age class collapses to near-zero F1 for every system: in this corpus, age-labelled tweets are frequently recollections of school bullying in which age is incidental rather than the target, so both the model and the lexicon route them to other_cyberbullying. Religion is also weak, being often implicit, whereas gender and ethnicity — carried by explicit slurs — score highest. The hybrid's gains concentrate exactly in the explicit-marker classes, consistent with its arbitration mechanism.

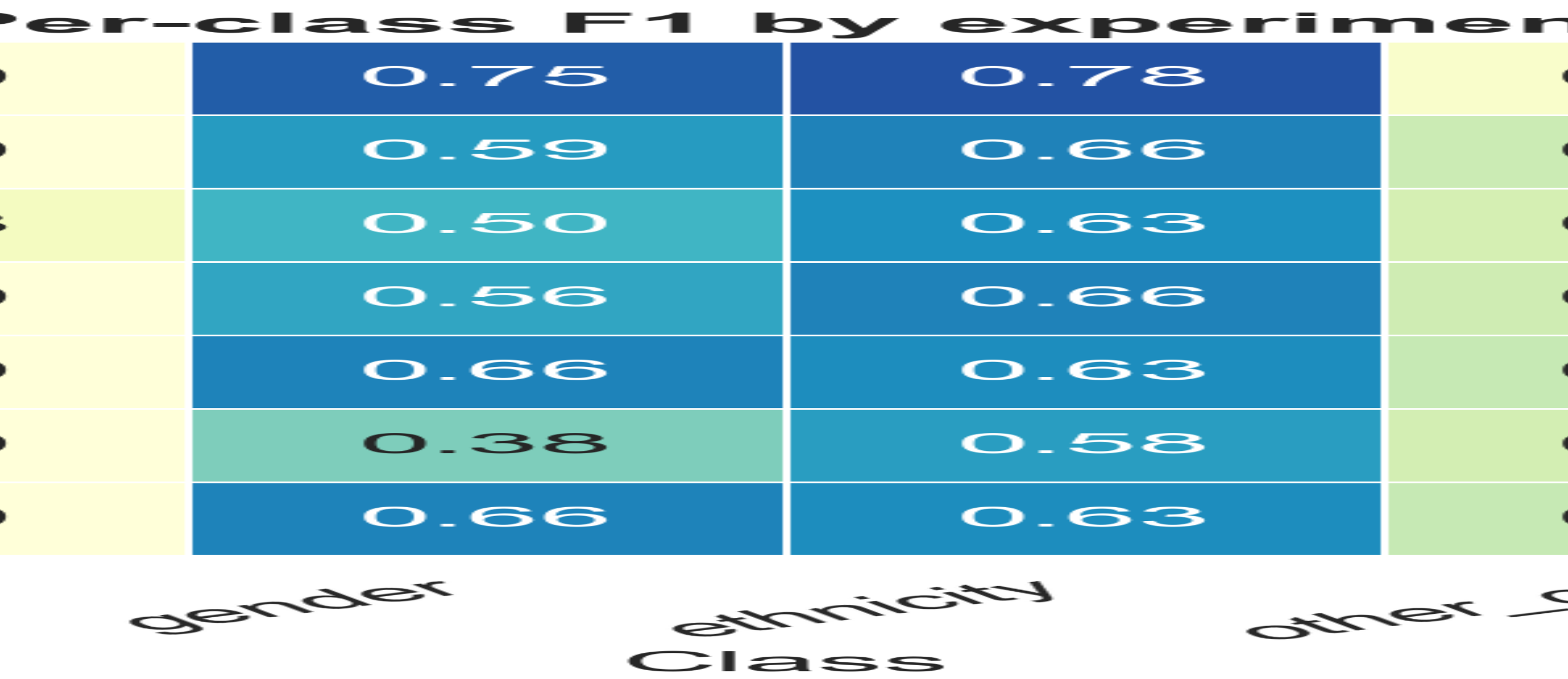

Figure 5. Per-class F1 heatmap across all systems.

Figure 6. Normalised confusion matrix of the hybrid system.

4.4 Hybrid Analysis

Built on the persona base, category arbitration applied 48 overrides, of which 33 were correct rescues and only 7 were breaks. This lifted accuracy from 0.400 to 0.443 (a gain of 0.043) and macro-F1 to 0.404. The empirical justification is decisive: on the cases where the lexicon fired a category and disagreed with the model, the lexicon was correct 45 times against the model's 8. Crucially, an attempt to use the lexicon instead as a recall safety-net produced almost no benefit, because the model's false negatives are implicit tweets (mean toxicity 0.04) that contain no lexicon terms — an honest negative result that motivated the arbitration design.

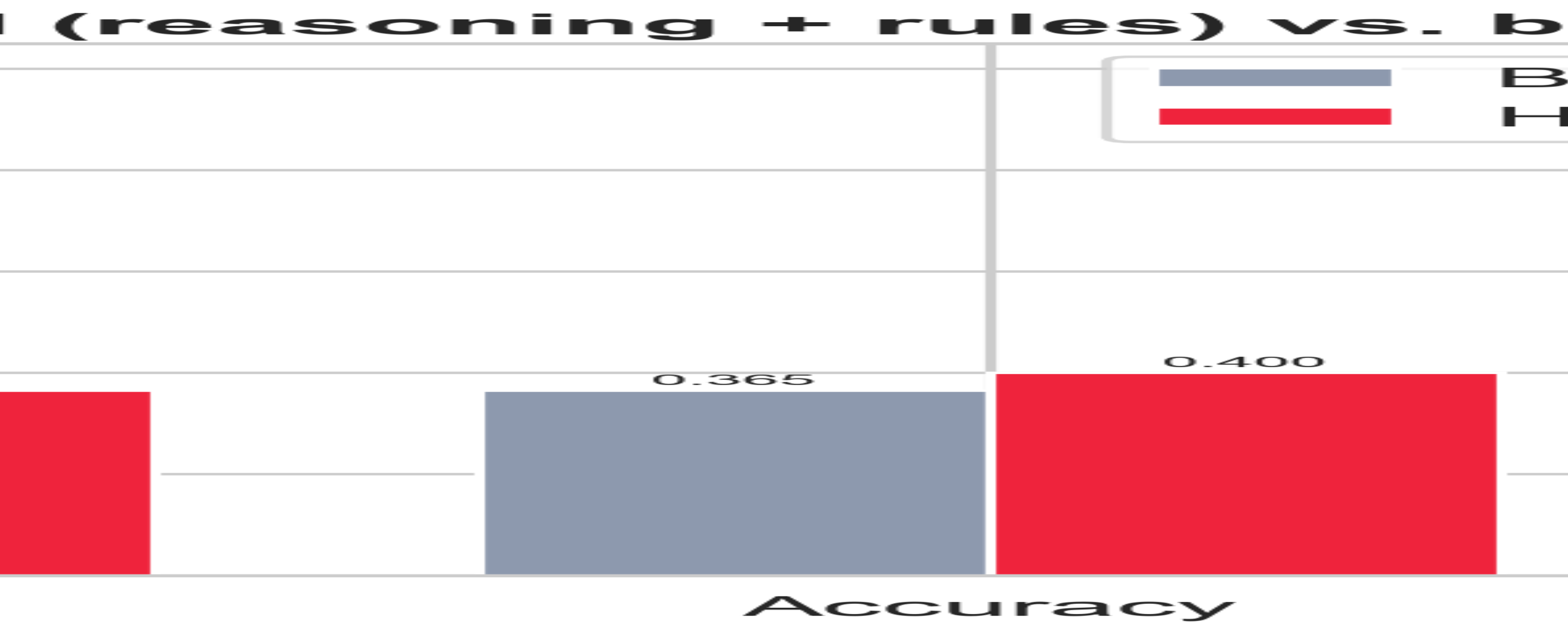

Figure 7. Hybrid versus the best pure model.

4.5 Robustness on Hard Cases

Prompt style	n hard	Acc. hard	Acc. easy	Acc. overall
zero_shot	123	0.431	0.371	0.383
few_shot	123	0.309	0.380	0.365
cot	123	0.358	0.352	0.353
persona	123	0.447	0.388	0.400
few_shot_cot	123	0.252	0.340	0.322

Table 7. Accuracy on hard (sarcastic / coded / quoted) versus easy tweets.

Persona is the only style that is *more* accurate on hard tweets than on easy ones, suggesting that role-conditioning helps the model slow down on ambiguous content; few-shot+CoT degrades most, mirroring its weak overall result.

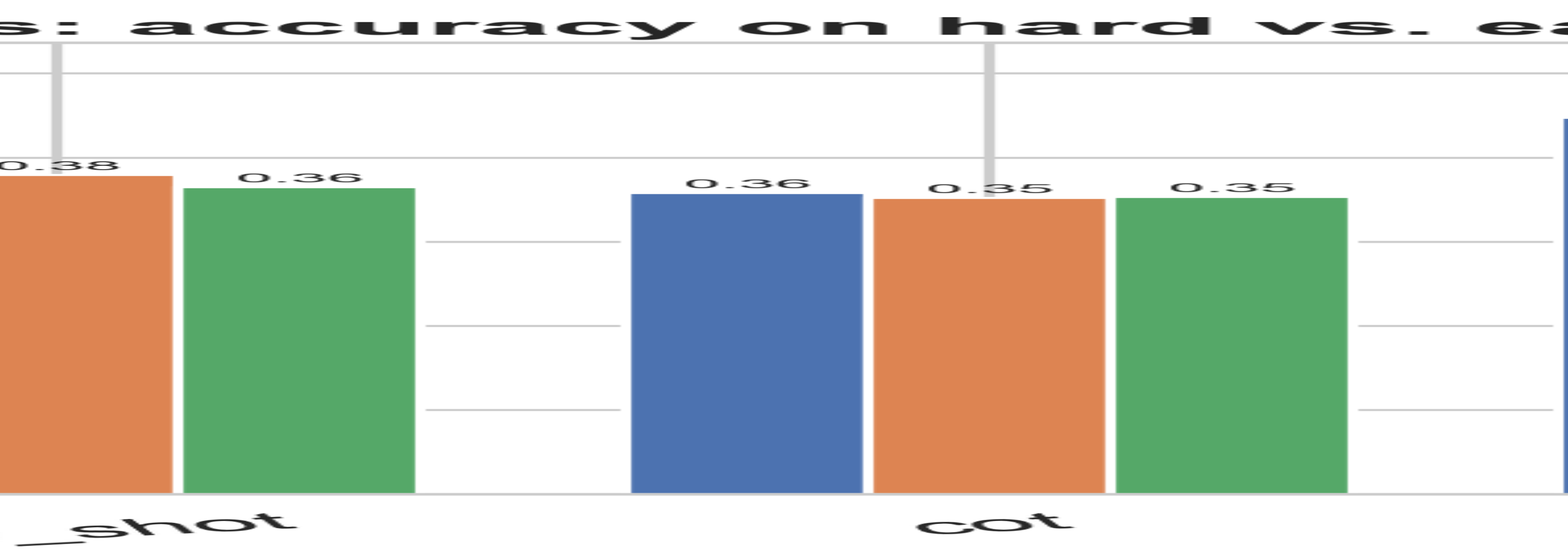

Figure 8. Accuracy on hard versus easy tweets by prompt style.

4.5.1 Failure Gallery

Table 8 lists representative hard errors. The pattern is consistent: implicit attacks on religion or age, and quoted or reported slurs, are routed to other_cyberbullying or to the wrong protected trait, because the abuse is conveyed by context rather than an explicit category marker.

Tweet (truncated)	Gold	Predicted
Every friend I had in middle school and high school were only my “friends” because they couldn’t bully me and I wasn’t scared of them and the other ha…	age	other_cyberbullying
I hope Twitter rips you a new one. You make me sick. @jimboslice_13 @ry_casper "how to discribe rape. His dick was hungry!" #notsexist	gender	other_cyberbullying
In high school I used to get bullied by a special ed girl bitch would dead ass hit me, and like fuck was I supposed to do	age	other_cyberbullying
@ryloxify PDP being careful. He's gone from rape jokes since EVERYBODY called him out of it to expanding his vocab to words like Gay & Nigga	gender	other_cyberbullying
Whoever is "their" in your mentioned tweet are jerks, a whole green flag doesn't look good to my eyes. Minorities are our backbone. Stay strong don't …	religion	other_cyberbullying
the second reminds me of the girl who bullied me in junior high and I’m just waiting for you to call me a bitch or trip me in gym class so depends on …	age	gender

Table 8. Representative hard-case misclassifications (chain-of-thought).

4.6 Explanation Quality

Rubric criterion	Mean score (0–1)
Composite explanation quality	0.569
Names the target group	0.308
Cites the reason / offending words	0.550
Describes intent coherently	0.950
Consistent with the gold label	0.467

Table 9. Explanation-quality rubric scores (LLM-as-judge, persona, n=120).

Explanations are coherent about intent (0.95) but frequently fail to name the precise target group (0.31), and consistency with the gold label (0.47) tracks the six-class accuracy. Quality is highest for gender and ethnicity and lowest for religion and age, exactly mirroring the classification confusions and confirming that the model’s uncertainty is reflected honestly in its explanations.

Chapter 5. Conclusion and Future Work

5.1 Summary of Findings

- **The hybrid is best.** ‘Reasoning + Rules’ via category arbitration beats the best pure model (+0.043 accuracy) and the rules alone (+0.027), reaching 0.443 accuracy and 0.404 macro-F1 while keeping 0.878 binary recall.
- **Arbitration, not a safety-net, is the correct fusion.** The model’s misses are implicit tweets the lexicon also misses; the lexicon’s value is precise category correction when explicit markers fire.
- **Role-conditioning beats reasoning scaffolds** on a small model; chain-of-thought and few-shot+CoT reduced macro-F1.
- **High recall, modest category precision.** The model catches roughly 88% of bullying but confuses which type, so binary moderation is dependable even when six-class accuracy is moderate.
- **Explanations are real but uneven**, strong on intent and weak on naming the exact target, with religion and age the hardest.

5.2 Limitations

The study uses one English-language corpus and one small model, and the evaluation sample is 600 tweets. The lexicon is intentionally compact, which bounds its recall. The dataset itself contains label noise — notably the age class — that caps the achievable six-class accuracy for any method. The LLM-as-judge for explanation quality, while consistent, is not a human study.

5.3 Future Work

- Scale the evaluation sample and add a larger model (for example a 70B variant) to separate prompt effects from model-capacity effects.
- Expand and weight the lexicon, and learn the arbitration threshold, to widen the hybrid’s coverage.
- Add a confidence-calibrated escalation path so uncertain cases are routed to human review.
- Conduct a human evaluation of explanation usefulness with teachers and moderators.
- Re-examine the age class definition, given its systematic confusion with general harassment.

5.4 Concluding Remarks

A carefully prompted Generative-AI model, supported by a small transparent rule layer, can act as a lightweight, explainable cyberbullying screen suitable for schools and platforms with limited compute. Its most dependable capability is binary safety detection, while fine-grained category labelling — and the precise naming of the targeted group in its explanations — is the clearest avenue for future improvement.